Lab Experiment-8

Aim

To implement a simple stochastic Part-of-Speech (POS) tagging algorithm using a basic probabilistic model to assign grammatical tags to words in a sentence.

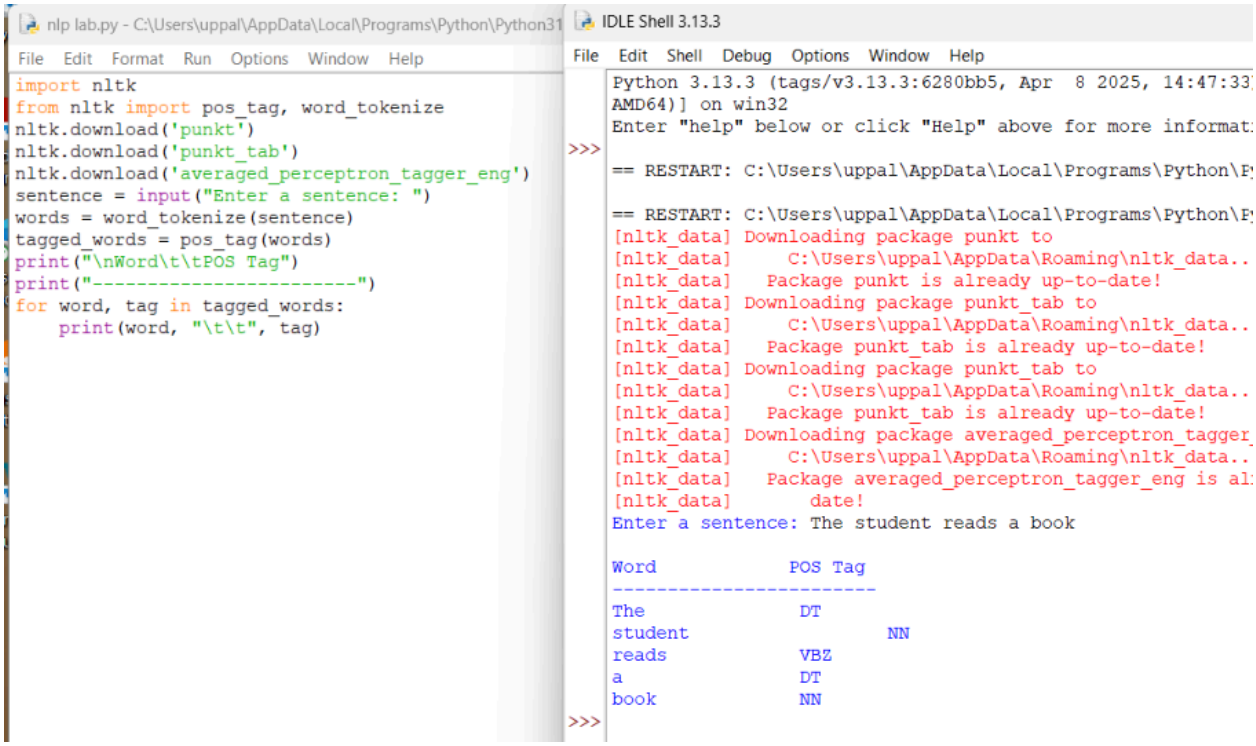
Algorithm

1. Start the program.
2. Import the pos_tag and word_tokenize functions from the NLTK library.
3. Read a sentence from the user.
4. Tokenize the sentence into individual words.
5. Apply the probabilistic POS tagging model to assign a grammatical tag to each word.
6. Display each word along with its corresponding POS tag.
7. Stop the program.

Code

```
import nltk
from nltk import pos_tag, word_tokenize
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('averaged_perceptron_tagger_eng')
sentence = input("Enter a sentence: ")
words = word_tokenize(sentence)
tagged_words = pos_tag(words)
print("\nWord\t\tPOS Tag")
print("------------------------")
for word, tag in tagged_words:
    print(word, "\t\t", tag)
```

Output:



Result

Thus, the stochastic POS tagging algorithm was implemented successfully in Python. The program tokenizes the input sentence and assigns appropriate Part-of-Speech tags to individual words using a probabilistic model.